

THE CONSTANT RANK THEOREM FOR COMPLEX LAGRANGIAN PHASE OPERATOR

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ABSTRACT. In this paper, we establish the constant rank theorem for the complex Lagrangian phase operator in both complex domains and Kähler manifolds. Following Li's work [21] on the constant rank property of the complex Hessian equation, we provide a new example of significant complex geometric equations that exhibit this property.

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1. INTRODUCTION

The constant rank theorem describes the phenomenon that, under appropriate structural conditions, the Hessian of a convex solution to an elliptic partial differential equation (PDE) must have constant rank. This result was first introduced and developed in the 1980s by Caffarelli and Friedman [5], Singer, Wong, Yau and Yau [26], and Korevaar and Lewis [20] during their investigations into the convexity of solutions to semilinear elliptic equations. Over the past two decades, the constant rank theorem for fully nonlinear elliptic PDEs has garnered significant attention, with notable contributions due to Caffarelli, Guan and Ma [4], Bian and Guan [1, 2], Székelyhidi and Weinkove [27, 28], and Ogden and Yuan [24].

In differential geometry, particularly in the study of the Christoffel-Minkowski problem and prescribed Weingarten curvature problem, the seminal works of Guan, Ma, and their collaborators [11, 12, 10] have demonstrated that the constant rank theorem plays an essential role in proving the existence of strictly convex solutions to these curvature problems. In recent years, a number of authors have further developed constant rank theorems in the context of Christoffel-Minkowski type problems, that is, fully nonlinear PDEs involving the second fundamental form of hypersurfaces; see, for example [16, 4, 3, 23, 18, 6, 29] and the references therein.

Furthermore, the constant rank theorem has also been generalized to the complex setting and has found applications in Kähler manifolds, as seen in the works of Li [21], Guan, Li and Zhang [9], Guan and Phong [13], among others [14, 25, 17]. Importantly, Li [21]

studied the complex Hessian equation

$$\sigma_k(u_{i\bar{j}}(z)) = f(z)$$

on a domain Ω in $\mathbb{C}^n$ and showed that when $f(z)$ satisfies certain structural condition that $-\log f(z)$ is plurisubharmonic, the complex Hessian $\partial\bar{\partial}u = (u_{i\bar{j}})_{n \times n}$ has constant rank in Ω . Motivated by this, we aim to explore whether similar constant rank results hold for other significant complex fully nonlinear elliptic PDEs, such as those arising from complex geometry.

In this work, we focus on the constant rank property of the complex Lagrangian phase equation

$$(1.1) \quad \sum_{i=1}^n \arctan \lambda_i(z) = \Theta(z) \quad \text{in } \Omega \subset \mathbb{C}^n,$$

where $(\lambda_1, \lambda_2, \dots, \lambda_n)$ are the eigenvalues of the complex Hessian $\partial\bar{\partial}u$. This equation can be viewed as a complex extension of the Lagrangian mean curvature equation

$$\sum_{i=1}^n \arctan \lambda_i(x) = \Theta(x) \quad \text{in } \Omega \subset \mathbb{R}^n,$$

in which λ_i ($i = 1, 2, \dots, n$) represent the eigenvalues of the real Hessian matrix D^2u . The function $\Theta(x)$ serves as the potential function for the mean curvature of the Lagrangian submanifold $\{(x, \nabla u(x)) \in \mathbb{R}^n \times \mathbb{R}^n : x \in \Omega\}$, with the condition $|\Theta| < \frac{n\pi}{2}$. When the phase $\Theta(x)$ is constant, the equation is referred to as the special Lagrangian equation, which was introduced by Harvey and Lawson [15] in the study of calibrated geometries. On the other hand, the complex Lagrangian phase equation (1.1) is closely related to the deformed Hermitian-Yang-Mills (dHYM) equation, which arises in the context of holomorphic line bundles over compact Kähler manifolds. The study of the dHYM equation originated in [19], and the solvability for the supercritical phase case ($\frac{(n-2)\pi}{2} < \Theta(z) < \frac{n\pi}{2}$) was later established by Collins, Jacob and Yau [7], under the assumption of the existence of an appropriate subsolution. For a broader discussion on this topic, one may refer to [19, 7] and the references therein. Regarding the solvability of the Dirichlet problem for equation (1.1) in the supercritical phase case, Collins, Picard and Wu [8] established an existence result under the assumption that a subsolution is available.

The main result of this paper are stated as follows.

Theorem 1.1 (Constant rank theorem). *Let Ω be a domain in $\mathbb{C}^n$, and $\Theta : \Omega \rightarrow (0, \frac{\pi}{2})$ be a C^2 function. Assume that u is a C^3 plurisubharmonic solution of (1.1) and if $-\log \tan \Theta$ is plurisubharmonic, then the rank of complex Hessian matrix $\partial\bar{\partial}u$ is constant in Ω .*

Remark 1.2. *In our theorem, we impose the condition $\Theta \in (0, \frac{\pi}{2})$, while our ultimate goal is to extend the result to $\Theta \in (0, \frac{n\pi}{2})$. The challenge arises from the non-positivity of (3.13)*

in Section 3, which depends on Lemma 2.4. Moreover, when determining the maximum value on the right-hand side of (3.15), the condition $\Theta \in (0, \frac{\pi}{2})$ is always required.

Once we obtain the constant rank theorem on flat domains, it is natural to consider the case of complex manifolds. We can provide a similar constant rank result on Kähler manifolds.

Theorem 1.3 (Kähler case). *Let (M, g) be a compact Kähler manifold of complex dimension n with Kähler metric $(g_{i\bar{j}})$, and with the orthogonal holomorphic bisectional curvature of g , $R_{i\bar{i}j\bar{j}}$ being bounded from below by $r > 0$. Let $\Theta : M \rightarrow (0, \frac{\pi}{2})$. Assume that u is a solution to the following equation, with $W = (w_{i\bar{j}}) = (u_{i\bar{j}} + g_{i\bar{j}})$ non-negative definite,*

$$(1.2) \quad \sum_{i=1}^n \arctan \lambda_i(W) = \Theta(z) \quad \text{in } M,$$

if the matrix $(\log \tan \Theta)_{i\bar{j}} - r g_{i\bar{j}}$ is non-positive definite for every $z \in M$, then the rank of W is constant in M .

Remark 1.4. *Compared to the proof of Theorem 4.1 in Li [21], for equation (1.2), when $\Theta \in (0, \frac{\pi}{2})$, we find through a refined analysis that the proof of the constant rank theorem for (1.2) can be obtained without relying on the homogeneity of elliptic operator.*

The organization of this paper is as follows. In Section 2, we provide some preliminary results, including elementary symmetric functions, computations related to the complex Lagrangian phase operator and so on. In Section 3, we establish a differential inequality (Lemma 3.1), which combined with the strong maximum principle, leads to the proof of Theorem 1.1. Section 4 is devoted to proving the constant rank result on Kähler manifolds, namely Theorem 1.3. Finally, in the Appendix A, we supplement the details of Case 3 in Lemma 3.1.

2. PRELIMINARIES

We recall the definition of k -th elementary symmetric functions. For $1 \leq k \leq n$ and $\lambda = (\lambda_1, \lambda_2, \dots, \lambda_n) \in \mathbb{C}^n$, we define

$$\sigma_k(\lambda) = \sum_{1 \leq i_1 < i_2 < \dots < i_k \leq n} \lambda_{i_1} \lambda_{i_2} \dots \lambda_{i_k}.$$

We also make the convention that $\sigma_0(\lambda) = 1$ and $\sigma_{-1}(\lambda) = \sigma_{n+1}(\lambda) = 0$. The definition of σ_k for vectors in $\mathbb{C}^n$ can be extended to $n \times n$ symmetric matrices in a natural way by letting $\sigma_k(W) = \sigma_k(\lambda(W))$, where $\lambda(W) = (\lambda_1(W), \lambda_2(W), \dots, \lambda_n(W))$ are the eigenvalues of the symmetric matrix W . We denote by $\sigma_{k-1}(W|i)$ the $(k-1)$ -th elementary symmetric function with $\lambda_i = 0$ and $\sigma_{k-2}(W|ij)$ ($i \neq j$) the $(k-2)$ -th elementary symmetric function with $\lambda_i = \lambda_j = 0$.

The first and second derivatives of the elementary symmetric functions are listed in the following lemma.

Lemma 2.1. *Suppose that $W = (W_{\alpha\bar{\beta}})$ is diagonal. For $k = 1, 2, \dots, n$, then we have*

$$(2.1) \quad \frac{\partial \sigma_k(W)}{\partial W_{\alpha\bar{\beta}}} = \begin{cases} \sigma_{k-1}(W|\alpha) & \text{if } \alpha = \beta, \\ 0 & \text{if } \alpha \neq \beta, \end{cases}$$

and

$$\frac{\partial^2 \sigma_k(W)}{\partial W_{\alpha\bar{\beta}} \partial W_{\gamma\bar{\delta}}} = \begin{cases} \sigma_{k-2}(W|\alpha\gamma) & \text{if } \alpha = \beta, \gamma = \delta, \alpha \neq \gamma, \\ -\sigma_{k-2}(W|\alpha\gamma) & \text{if } \alpha = \delta, \beta = \gamma, \alpha \neq \gamma, \\ 0 & \text{otherwise.} \end{cases}$$

Newton's inequality for the k -th elementary symmetric functions is a fundamental result, which will be used in the next section. For completeness, we state it below. A proof can be found in [22].

Lemma 2.2 (Newton's inequality). *For any $k \in \{1, 2, \dots, n-1\}$, and $\lambda \in \mathbb{C}^n$, we have*

$$(n-k+1)(k+1)\sigma_{k-1}(\lambda)\sigma_{k+1}(\lambda) \leq k(n-k)\sigma_k^2(\lambda).$$

In particular, we get

$$\frac{\sigma_{k+1}(\lambda)}{\sigma_k(\lambda)} \leq \frac{\sigma_k(\lambda)}{\sigma_{k-1}(\lambda)}.$$

Let F be the complex Lagrangian phase operator, defined by

$$F(W) = \sum_{i=1}^n \arctan \lambda_i(W),$$

where $W = (W_{\alpha\bar{\beta}})$ is a Hermitian matrix and $\lambda_i(W)$, $i = 1, 2, \dots, n$, are the eigenvalues of W . The first and second order derivatives of F can be explicitly computed as follows.

Lemma 2.3. *Suppose that W is diagonal, we have*

$$F^{\alpha\bar{\beta}} = \frac{\partial F(W)}{\partial W_{\alpha\bar{\beta}}} = \begin{cases} \frac{1}{1+\lambda_\alpha^2} & \text{if } \alpha = \beta, \\ 0 & \text{if } \alpha \neq \beta, \end{cases}$$

and

$$F^{\alpha\bar{\beta}, \gamma\bar{\delta}} = \frac{\partial^2 F(W)}{\partial W_{\alpha\bar{\beta}} \partial W_{\gamma\bar{\delta}}} = \begin{cases} -\frac{2\lambda_\alpha}{(1+\lambda_\alpha^2)^2} & \text{if } \alpha = \beta = \gamma = \delta, \\ -\frac{\lambda_\alpha + \lambda_\beta}{(1+\lambda_\alpha^2)(1+\lambda_\beta^2)} & \text{if } \alpha = \delta, \beta = \gamma, \\ 0 & \text{otherwise.} \end{cases}$$

We now turn to two fundamental results, whose proofs are included for completeness.

Lemma 2.4. *For $x, y \in \mathbb{R}$, set $\alpha = \arctan x$, $\beta = \arctan y$ with $\alpha, \beta > 0$. When $\alpha + \beta \in (0, \frac{\pi}{2})$, it follows that $xy < 1$.*

Proof. Since $\alpha \in (0, \frac{\pi}{2} - \beta)$ and $\tan$ is a strictly increasing function, we have $\tan \alpha < \tan(\frac{\pi}{2} - \beta)$. This means that $x < \cot \beta = \frac{1}{y}$, which leads to $xy < 1$. $\square$

Lemma 2.5. *Consider the following n -th order matrix*

$$M_n = \begin{pmatrix} a_1 & b & \cdots & b & b \\ b & a_2 & \cdots & b & b \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ b & b & \cdots & a_{n-1} & b \\ b & b & \cdots & b & a_n \end{pmatrix},$$

then its determinant is given by

$$\det M_n = \prod_{i=1}^n (a_i - b) + b \sum_{i=1}^n \prod_{j \neq i} (a_j - b).$$

Proof. To derive the recurrence relation, we split the n -th column of the matrix M_n , which gives us

$$\begin{aligned} \det M_n &= \det \begin{pmatrix} a_1 & b & \cdots & b & b \\ b & a_2 & \cdots & b & b \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ b & b & \cdots & a_{n-1} & b \\ b & b & \cdots & b & b \end{pmatrix} + \det \begin{pmatrix} a_1 & b & \cdots & b & 0 \\ b & a_2 & \cdots & b & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ b & b & \cdots & a_{n-1} & 0 \\ b & b & \cdots & b & a_n - b \end{pmatrix} \\ &= \det \begin{pmatrix} a_1 - b & 0 & \cdots & 0 & 0 \\ 0 & a_2 - b & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & a_{n-1} - b & 0 \\ b & b & \cdots & b & a_n \end{pmatrix} + (a_n - b) \det M_{n-1} \\ &= b \prod_{i=1}^{n-1} (a_i - b) + (a_n - b) \det M_{n-1}. \end{aligned}$$

Then we conclude that

$$\det M_n = \prod_{i=1}^n (a_i - b) + b \sum_{i=1}^n \prod_{j \neq i} (a_j - b).$$

$\square$

Finally, we list some preliminary knowledge on Kähler geometry. Let (M, g, J) be a compact Kähler manifold with the Kähler metric g . In local coordinates $(z_1, z_2, \dots, z_n)$, we have a local basis $\{\frac{\partial}{\partial z_1}, \frac{\partial}{\partial z_2}, \dots, \frac{\partial}{\partial z_n}\}$ for holomorphic tangent bundle of M . Since the complex structure J is parallel, the only nonzero terms of Christoffel symbols are Γ_{ij}^k and

$\Gamma_{i\bar{j}}^{\bar{k}}$. If $g_{i\bar{j}} = g\left(\frac{\partial}{\partial z_i}, \frac{\partial}{\partial \bar{z}_j}\right)$ denote the metric tensor in local coordinates, then

$$\Gamma_{ij}^k = \sum_l g^{k\bar{l}} \frac{\partial g_{i\bar{l}}}{\partial z_j}, \quad \Gamma_{i\bar{j}}^{\bar{k}} = \sum_l g^{l\bar{k}} \frac{\partial g_{l\bar{i}}}{\partial \bar{z}_j},$$

where $g^{i\bar{j}}$ are the components of the inverse of g . The curvature tensor of the metric $g_{i\bar{j}}$ is defined as

$$R_{i\bar{j}k\bar{l}} = \sum_p g_{p\bar{j}} R_{ik\bar{l}}^p = -\frac{\partial^2 g_{i\bar{j}}}{\partial z_k \partial \bar{z}_l} + \sum_{p,q} g^{p\bar{q}} \frac{\partial g_{i\bar{q}}}{\partial z_k} \frac{\partial g_{p\bar{j}}}{\partial \bar{z}_l}.$$

By the Kähler condition, we have

$$R_{i\bar{j}k\bar{l}} = R_{k\bar{j}i\bar{l}} = R_{i\bar{l}k\bar{j}} = R_{k\bar{l}i\bar{j}}.$$

We say that (M, g, J) has positive holomorphic bisectional curvature if

$$\sum_{i,j,k,l} R_{i\bar{j}k\bar{l}} v^i v^{\bar{j}} w^k w^{\bar{l}} > 0$$

for all nonzero holomorphic tangent vectors v and w . On a compact Kähler manifold, the orthogonal bisectional curvature is bounded below by r means that $R_{i\bar{i}j\bar{j}} \geq r$ for any $i \neq j$.

In local coordinates, given by $(1, 1)$ tensor $T_{i\bar{j}}$, its covariant derivatives are defined as

$$T_{i\bar{j},k} = \frac{\partial T_{i\bar{j}}}{\partial z_k} - \sum_l \Gamma_{ik}^l T_{l\bar{j}}$$

and

$$T_{i\bar{j},\bar{k}} = \frac{\partial T_{i\bar{j}}}{\partial \bar{z}_k} - \sum_l \Gamma_{j\bar{k}}^{\bar{l}} T_{i\bar{l}}.$$

We present the commutation formulas for covariant derivatives that will be used later,

$$\begin{aligned} T_{i\bar{j},kl} &= T_{i\bar{j},lk}, \\ T_{i\bar{j},\bar{k}\bar{l}} &= T_{i\bar{j},\bar{l}\bar{k}}, \\ T_{i\bar{j},k\bar{l}} &= T_{i\bar{j},\bar{l}k} + \sum_{p,q} g^{p\bar{q}} R_{i\bar{q}k\bar{l}} T_{p\bar{j}} - \sum_{p,q} g^{p\bar{q}} R_{p\bar{j}k\bar{l}} T_{i\bar{q}}. \end{aligned}$$

3. THE CASE OF COMPLEX DOMAINS

This section is devoted to the proof of Theorem 1.1. The main idea is to establish a differential inequality near the point where the complex Hessian matrix $\partial\bar{\partial}u = (u_{i\bar{j}})$ attains its minimal rank, as stated in the following Lemma 3.1. Then, using the strong maximum principle, we will complete the proof of the theorem. Assume that the complex Hessian $\partial\bar{\partial}u$ attains its minimal rank l at some points $z_0 \in \Omega$. We select an open neighborhood $\mathcal{O}$ of z_0 and define

$$\phi(z) = \sigma_{l+1}(u_{i\bar{j}}(z)) \quad \text{in } \mathcal{O}.$$

Lemma 3.1. *Let $\Omega \subset \mathbb{C}^n$ be any domains, $\Theta : \Omega \rightarrow (0, \frac{\pi}{2})$ be a C^2 function, and $u \in C^3(\Omega)$ be a plurisubharmonic solution of (1.1). If the complex Hessian $\partial\bar{\partial}u$ attains its minimal rank l at $z_0 \in \Omega$, and $\mathcal{O}$ is a neighborhood of z_0 , then there exists positive constant C , depending only on u , Θ and n , such that*

$$(3.1) \quad \sum_{\alpha, \beta} F^{\alpha\bar{\beta}} \phi_{\alpha\bar{\beta}} \leq C(|\nabla\phi| + \phi) \quad \text{in } \mathcal{O}.$$

Proof. For any $z \in \mathcal{O}$, let $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n$ be the eigenvalues of $\partial\bar{\partial}u$ at z . There is a positive constant $c > 0$ such that $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_l \geq c$. Let $G = \{1, 2, \dots, l\}$ and $B = \{l+1, l+2, \dots, n\}$ be the “good” and “bad” sets of indices, respectively. If there is no confusion, we also denote the “good” and “bad” eigenvalues of $\partial\bar{\partial}u$ by G and B , respectively.

Following the notations in [5], for two functions $h(z)$ and $k(z)$ defined in $\mathcal{O}$, we say that $h(z) \lesssim k(z)$ if there exists a positive constant C_1 such that

$$h(z) - k(z) \leq C_1(|\nabla\phi(z)| + \phi(z)).$$

We also write $h(z) \sim k(z)$ if $h(z) \lesssim k(z)$ and $k(z) \lesssim h(z)$. Next, we write $h \lesssim k$ if the above inequality holds for any $z \in \mathcal{O}$, with constant C_1 independent of z . Finally, $h \sim k$ means $h \lesssim k$ and $k \lesssim h$.

For each $z \in \mathcal{O}$ fixed, by choosing orthonormal complex coordinates, we suppose that the complex Hessian $\partial\bar{\partial}u$ is diagonal at z and $u_{i\bar{i}} = \lambda_i$ for any $i = 1, 2, \dots, n$. From now on, all calculations will be carried out at the fixed point z . When we use the relation $\lesssim$, all constants involved are under controlled.

Since $\partial\bar{\partial}u$ is diagonal, we have

$$0 \sim \phi \sim \sigma_l(G) \sum_{i \in B} u_{i\bar{i}} \sim \sum_{i \in B} u_{i\bar{i}},$$

and

$$(3.2) \quad u_{i\bar{i}} \sim 0 \quad i \in B.$$

The relation (3.2) yields

$$(3.3) \quad \sigma_l(\lambda|i) \sim \begin{cases} 0 & \text{if } i \in G, \\ \sigma_l(G) & \text{if } i \in B, \end{cases}$$

and

$$(3.4) \quad \sigma_{l-1}(\lambda|ij) \sim \begin{cases} 0 & \text{if } i \in G, j \in G, \\ \sigma_{l-1}(G|i) & \text{if } i \in G, j \in B, \\ \sigma_{l-1}(G|j) & \text{if } i \in B, j \in G, \\ \sigma_{l-1}(G) & \text{if } i \in B, j \in B. \end{cases}$$

Taking the first derivatives of ϕ , by (3.3) and (2.1), we get

$$0 \sim \phi_\alpha = \sum_{i,j} \frac{\partial \sigma_{l+1}}{\partial u_{i\bar{j}}} u_{i\bar{j}\alpha} = \sum_i \sigma_l(\lambda|i) u_{i\bar{i}\alpha} \sim \sigma_l(G) \sum_{i \in B} u_{i\bar{i}\alpha},$$

namely,

$$(3.5) \quad \sum_{i \in B} u_{i\bar{i}\alpha} \sim 0.$$

Similarly,

$$(3.6) \quad \sum_{i \in B} u_{i\bar{i}\bar{\alpha}} \sim 0.$$

Using relations (3.3)-(3.6) and Lemma 2.1, for $\alpha = 1, 2, \dots, n$, we calculate the second derivatives of ϕ as

$$\begin{aligned} \phi_{\alpha\bar{\alpha}} &= \sum_{i,j} \frac{\partial \sigma_{l+1}}{\partial u_{i\bar{j}}} u_{i\bar{j}\alpha\bar{\alpha}} + \sum_{i,j,p,q} \frac{\partial^2 \sigma_{l+1}}{\partial u_{i\bar{j}} \partial u_{p\bar{q}}} u_{i\bar{j}\alpha} u_{p\bar{q}\bar{\alpha}} \\ &= \sum_i \sigma_l(\lambda|i) u_{i\bar{i}\alpha\bar{\alpha}} + \sum_{i \neq j} \sigma_{l-1}(\lambda|ij) u_{i\bar{i}\alpha} u_{j\bar{j}\bar{\alpha}} - \sum_{i \neq j} \sigma_{l-1}(\lambda|ij) u_{i\bar{j}\alpha} u_{j\bar{i}\bar{\alpha}} \\ &\sim \sigma_l(G) \sum_{i \in B} u_{i\bar{i}\alpha\bar{\alpha}} + \left(\sum_{\substack{i \in G \\ j \in B}} + \sum_{\substack{i \in B \\ j \in G}} + \sum_{\substack{i,j \in G \\ i \neq j}} + \sum_{\substack{i,j \in B \\ i \neq j}} \right) \sigma_{l-1}(\lambda|ij) u_{i\bar{i}\alpha} u_{j\bar{j}\bar{\alpha}} \\ &\quad - \left(\sum_{\substack{i \in G \\ j \in B}} + \sum_{\substack{i \in B \\ j \in G}} + \sum_{\substack{i,j \in G \\ i \neq j}} + \sum_{\substack{i,j \in B \\ i \neq j}} \right) \sigma_{l-1}(\lambda|ij) |u_{i\bar{j}\alpha}|^2 \\ &\sim \sigma_l(G) \sum_{i \in B} u_{i\bar{i}\alpha\bar{\alpha}} + \sum_{i \in G} \sigma_{l-1}(G|i) u_{i\bar{i}\alpha} \sum_{j \in B} u_{j\bar{j}\bar{\alpha}} + \sum_{i \in B} u_{i\bar{i}\alpha} \sum_{j \in G} \sigma_{l-1}(G|j) u_{j\bar{j}\bar{\alpha}} \\ &\quad + \sigma_{l-1}(G) \sum_{\substack{i,j \in B \\ i \neq j}} u_{i\bar{i}\alpha} u_{j\bar{j}\bar{\alpha}} - \sum_{\substack{i \in G \\ j \in B}} \sigma_{l-1}(G|i) (|u_{i\bar{j}\alpha}|^2 + |u_{j\bar{i}\alpha}|^2) - \sigma_{l-1}(G) \sum_{\substack{i,j \in B \\ i \neq j}} |u_{i\bar{j}\alpha}|^2 \\ &\sim \sigma_l(G) \sum_{i \in B} u_{i\bar{i}\alpha\bar{\alpha}} + \sigma_{l-1}(G) \sum_{\substack{i,j \in B \\ i \neq j}} u_{i\bar{i}\alpha} u_{j\bar{j}\bar{\alpha}} - \sum_{\substack{i \in G \\ j \in B}} \sigma_{l-1}(G|i) (|u_{i\bar{j}\alpha}|^2 + |u_{j\bar{i}\alpha}|^2) \\ &\quad - \sigma_{l-1}(G) \sum_{\substack{i,j \in B \\ i \neq j}} |u_{i\bar{j}\alpha}|^2. \end{aligned}$$

From (3.6), we get $-u_{i\bar{i}\bar{\alpha}} \sim \sum_{\substack{j \in B \\ j \neq i}} u_{j\bar{j}\bar{\alpha}}$, which implies

$$\sigma_{l-1}(G) \sum_{\substack{i,j \in B \\ i \neq j}} u_{i\bar{i}\alpha} u_{j\bar{j}\bar{\alpha}} \sim -\sigma_{l-1}(G) \sum_{i \in B} |u_{i\bar{i}\alpha}|^2.$$

Hence, we derive

$$\phi_{\alpha\bar{\alpha}} \sim \sigma_l(G) \sum_{i \in B} u_{i\bar{i}\alpha\bar{\alpha}} - \sigma_{l-1}(G) \sum_{i,j \in B} |u_{i\bar{j}\alpha}|^2 - \sum_{\substack{i \in G \\ j \in B}} \sigma_{l-1}(G|i|)(|u_{i\bar{j}\alpha}|^2 + |u_{j\bar{i}\alpha}|^2).$$

By Lemma 2.3, we know that $F^{\alpha\bar{\beta}}$ is diagonal at z , therefore

$$\begin{aligned} \sum_{\alpha, \beta} F^{\alpha\bar{\beta}} \phi_{\alpha\bar{\beta}} &= \sum_{\alpha} F^{\alpha\bar{\alpha}} \phi_{\alpha\bar{\alpha}} \\ (3.7) \quad &\sim \sigma_l(G) \sum_{\alpha} \sum_{i \in B} F^{\alpha\bar{\alpha}} u_{i\bar{i}\alpha\bar{\alpha}} - \sigma_{l-1}(G) \sum_{\alpha} \sum_{i,j \in B} F^{\alpha\bar{\alpha}} |u_{i\bar{j}\alpha}|^2 \\ &\quad - \sum_{\alpha} \sum_{\substack{i \in G \\ j \in B}} \sigma_{l-1}(G|i|) F^{\alpha\bar{\alpha}} (|u_{i\bar{j}\alpha}|^2 + |u_{j\bar{i}\alpha}|^2). \end{aligned}$$

To begin with, we need to handle the fourth-order derivative terms appearing in (3.7). Taking the first-order derivative of both sides of the equation $F(\partial\bar{\partial}u) = \Theta(z)$ with respect to z_i and $\bar{z}_i$, we obtain

$$(3.8) \quad \sum_{\alpha} F^{\alpha\bar{\alpha}} u_{\alpha\bar{\alpha}i} = \Theta_i \quad \text{and} \quad \sum_{\alpha} F^{\alpha\bar{\alpha}} u_{\alpha\bar{\alpha}\bar{i}} = \Theta_{\bar{i}}.$$

Differentiating (3.8) once more yields

$$(3.9) \quad \sum_{\alpha, \beta, \gamma, \delta} F^{\alpha\bar{\beta}, \gamma\bar{\delta}} u_{\alpha\bar{\beta}i} u_{\gamma\bar{\delta}\bar{i}} + \sum_{\alpha, \beta} F^{\alpha\bar{\beta}} u_{\alpha\bar{\beta}i\bar{i}} = \Theta_{i\bar{i}}.$$

According to Lemma 2.3, it follows from (3.9) that

$$(3.10) \quad \sum_{\alpha} F^{\alpha\bar{\alpha}} u_{\alpha\bar{\alpha}i\bar{i}} = - \sum_{\alpha} F^{\alpha\bar{\alpha}, \alpha\bar{\alpha}} |u_{\alpha\bar{\alpha}i}|^2 - \sum_{\alpha \neq \beta} F^{\alpha\bar{\beta}, \beta\bar{\alpha}} |u_{\alpha\bar{\beta}i}|^2 + \Theta_{i\bar{i}}.$$

Substituting (3.10) into (3.7), we arrive at

$$\begin{aligned} \sum_{\alpha} F^{\alpha\bar{\alpha}} \phi_{\alpha\bar{\alpha}} &\sim - \sigma_l(G) \sum_{\alpha} \sum_{i \in B} F^{\alpha\bar{\alpha}, \alpha\bar{\alpha}} |u_{\alpha\bar{\alpha}i}|^2 - \sigma_l(G) \sum_{\alpha \neq \beta} \sum_{i \in B} F^{\alpha\bar{\beta}, \beta\bar{\alpha}} |u_{\alpha\bar{\beta}i}|^2 \\ (3.11) \quad &\quad - \sum_{\alpha} \sum_{\substack{i \in G \\ j \in B}} \sigma_{l-1}(G|i|) F^{\alpha\bar{\alpha}} (|u_{i\bar{j}\alpha}|^2 + |u_{j\bar{i}\alpha}|^2) \\ &\quad - \sigma_{l-1}(G) \sum_{\alpha} \sum_{i,j \in B} F^{\alpha\bar{\alpha}} |u_{i\bar{j}\alpha}|^2 + \sigma_l(G) \sum_{i \in B} \Theta_{i\bar{i}}. \end{aligned}$$

In what follows, we need to carefully analyze the third-order derivative terms of u on the right-hand side of (3.11), specifically the first four terms, which we denote as ①–④. Clearly, the term ④ is non-positive. We now analyze terms ①–③. For the term ②, Lemma 2.3 implies

$$\textcircled{2} = - \sigma_l(G) \sum_{\alpha \neq \beta} \sum_{i \in B} F^{\alpha\bar{\beta}, \beta\bar{\alpha}} |u_{\alpha\bar{\beta}i}|^2$$

$$\begin{aligned}
&= -\sigma_l(G) \left(\sum_{\substack{\alpha \in G \\ \beta \in B}} + \sum_{\substack{\alpha \in B \\ \beta \in G}} + \sum_{\substack{\alpha, \beta \in G \\ \alpha \neq \beta}} + \sum_{\substack{\alpha, \beta \in B \\ \alpha \neq \beta}} \right) \sum_{i \in B} F^{\alpha\bar{\beta}, \beta\bar{\alpha}} |u_{\alpha\bar{\beta}i}|^2 \\
&\sim \sigma_l(G) \sum_{\substack{\alpha \in G \\ \beta \in B}} \sum_{i \in B} \frac{\lambda_\alpha}{1 + \lambda_\alpha^2} (|u_{\alpha\bar{\beta}i}|^2 + |u_{\beta\bar{\alpha}i}|^2) \\
&\quad + \sigma_l(G) \sum_{\substack{\alpha, \beta \in G \\ \alpha \neq \beta}} \sum_{i \in B} \frac{\lambda_\alpha + \lambda_\beta}{(1 + \lambda_\alpha^2)(1 + \lambda_\beta^2)} |u_{\alpha\bar{\beta}i}|^2 \\
&\stackrel{\triangle}{=} \textcircled{2}_1 + \textcircled{2}_2.
\end{aligned}$$

Further, we analyze term $\textcircled{3}$ and obtain

$$\begin{aligned}
\textcircled{3} &= - \sum_{\alpha} \sum_{\substack{i \in G \\ j \in B}} \sigma_{l-1}(G|i) F^{\alpha\bar{\alpha}} (|u_{i\bar{j}\alpha}|^2 + |u_{j\bar{i}\alpha}|^2) \\
&\sim - \sum_{\alpha \in G} \sum_{\substack{i \in G \\ j \in B}} \sigma_{l-1}(G|i) \frac{1}{1 + \lambda_\alpha^2} (|u_{i\bar{j}\alpha}|^2 + |u_{j\bar{i}\alpha}|^2) \\
&\quad - \sum_{\alpha \in B} \sum_{\substack{i \in G \\ j \in B}} \sigma_{l-1}(G|i) (|u_{i\bar{j}\alpha}|^2 + |u_{j\bar{i}\alpha}|^2) \\
&= - \sum_{i \in B} \sum_{\substack{\alpha, \beta \in G \\ \alpha \neq \beta}} \sigma_{l-1}(G|\beta) \frac{1}{1 + \lambda_\alpha^2} (|u_{\beta\bar{i}\alpha}|^2 + |u_{i\bar{\beta}\alpha}|^2) \\
&\quad - \sum_{i \in B} \sum_{\alpha \in G} \sigma_{l-1}(G|\alpha) \frac{1}{1 + \lambda_\alpha^2} (|u_{\alpha\bar{i}\alpha}|^2 + |u_{\alpha\bar{\alpha}i}|^2) \\
&\quad - \sum_{i \in B} \sum_{\substack{\alpha \in G \\ \beta \in B}} \sigma_{l-1}(G|\alpha) (|u_{\alpha\bar{\beta}i}|^2 + |u_{\beta\bar{\alpha}i}|^2) \\
&\stackrel{\triangle}{=} \textcircled{3}_1 + \textcircled{3}_2 + \textcircled{3}_3.
\end{aligned}$$

Combining $\textcircled{2}_1$ with $\textcircled{3}_3$, we get

$$\begin{aligned}
\textcircled{2}_1 + \textcircled{3}_3 &= \sum_{i \in B} \sum_{\substack{\alpha \in G \\ \beta \in B}} \left(\sigma_l(G) \frac{\lambda_\alpha}{1 + \lambda_\alpha^2} - \sigma_{l-1}(G|\alpha) \right) (|u_{\alpha\bar{\beta}i}|^2 + |u_{\beta\bar{\alpha}i}|^2) \\
(3.12) \quad &= -\sigma_l(G) \sum_{i \in B} \sum_{\substack{\alpha \in G \\ \beta \in B}} \frac{1}{\lambda_\alpha(1 + \lambda_\alpha^2)} (|u_{\alpha\bar{\beta}i}|^2 + |u_{\beta\bar{\alpha}i}|^2) \\
&\leq 0.
\end{aligned}$$

For the terms $\textcircled{2}_2$ and $\textcircled{3}_1$, by Lemma 2.4, we can conclude that

$$\begin{aligned}
(3.13) \quad \textcircled{2}_2 + \textcircled{3}_1 &= \sum_{i \in B} \sum_{\substack{\alpha, \beta \in G \\ \alpha \neq \beta}} \sigma_l(G) \frac{\lambda_\alpha + \lambda_\beta}{(1 + \lambda_\alpha^2)(1 + \lambda_\beta^2)} |u_{\alpha\bar{\beta}i}|^2 \\
&\quad - \sum_{i \in B} \sum_{\substack{\alpha, \beta \in G \\ \alpha \neq \beta}} \sigma_{l-1}(G|\beta) \frac{1}{1 + \lambda_\alpha^2} (|u_{\beta\bar{i}\alpha}|^2 + |u_{i\bar{\beta}\alpha}|^2) \\
&\leq \sum_{i \in B} \sum_{\substack{\alpha, \beta \in G \\ \alpha \neq \beta}} \left(\sigma_l(G) \frac{\lambda_\alpha + \lambda_\beta}{(1 + \lambda_\alpha^2)(1 + \lambda_\beta^2)} - \sigma_{l-1}(G|\beta) \frac{1}{1 + \lambda_\alpha^2} \right) |u_{\alpha\bar{\beta}i}|^2 \\
&= \sum_{i \in B} \sum_{\substack{\alpha, \beta \in G \\ \alpha \neq \beta}} \sigma_{l-1}(G|\beta) \frac{\lambda_\alpha \lambda_\beta - 1}{(1 + \lambda_\alpha^2)(1 + \lambda_\beta^2)} |u_{\alpha\bar{\beta}i}|^2 \\
&\leq 0.
\end{aligned}$$

Considering terms $\textcircled{1}$ and $\textcircled{3}_2$, and applying Lemma 2.3 again, we have

$$\begin{aligned}
(3.14) \quad \textcircled{1} + \textcircled{3}_2 &\sim \sigma_l(G) \sum_{i \in B} \sum_{\alpha \in G} \frac{2\lambda_\alpha}{(1 + \lambda_\alpha^2)^2} |u_{\alpha\bar{\alpha}i}|^2 \\
&\quad - \sigma_l(G) \sum_{i \in B} \sum_{\alpha \in G} \frac{1}{\lambda_\alpha(1 + \lambda_\alpha^2)} (|u_{\alpha\bar{i}\alpha}|^2 + |u_{\alpha\bar{\alpha}i}|^2) \\
&\leq \sigma_l(G) \sum_{i \in B} \sum_{\alpha \in G} \frac{\lambda_\alpha^2 - 1}{\lambda_\alpha(1 + \lambda_\alpha^2)^2} |u_{\alpha\bar{\alpha}i}|^2.
\end{aligned}$$

Thus, we can deduce from (3.12), (3.13) and (3.14) that

$$(3.15) \quad \frac{1}{\sigma_l(G)} \sum_{\alpha} F^{\alpha\bar{\alpha}} \phi_{\alpha\bar{\alpha}} \lesssim \sum_{i \in B} \sum_{\alpha \in G} \frac{\lambda_\alpha^2 - 1}{\lambda_\alpha(1 + \lambda_\alpha^2)^2} |u_{\alpha\bar{\alpha}i}|^2 + \sum_{i \in B} \Theta_{i\bar{i}}.$$

The next challenge is to find the maximum of the first term on the right-hand side (3.15) under constraints given by (3.8). We will overcome this difficulty by employing the method of Lagrange multipliers. This calculation is highly technical.

For simplicity of notations, fix $i \in B = \{l+1, l+2, \dots, n\}$, and for any $\alpha \in G = \{1, 2, \dots, l\}$, denote

$$\frac{1}{1 + \lambda_\alpha^2} u_{\alpha\bar{\alpha}i} = X_\alpha + \sqrt{-1}Y_\alpha, \quad \text{and} \quad \frac{1}{1 + \lambda_\alpha^2} u_{\alpha\bar{i}\alpha} = X_\alpha - \sqrt{-1}Y_\alpha,$$

where X_α and Y_α are real valued functions. We consider the following quadratic polynomial

$$(3.16) \quad f(X_1, X_2, \dots, X_l, Y_1, Y_2, \dots, Y_l) = \sum_{\alpha \in G} \frac{\lambda_\alpha^2 - 1}{\lambda_\alpha} (X_\alpha^2 + Y_\alpha^2),$$

subject to the constraints

$$\sum_{\alpha \in G} (X_\alpha + \sqrt{-1}Y_\alpha) \sim \Theta_i, \quad \text{and} \quad \sum_{\alpha \in G} (X_\alpha - \sqrt{-1}Y_\alpha) \sim \Theta_{\bar{i}}.$$

We use $\Re(\Theta_i)$ to denote the real part of Θ_i , and $\Im(\Theta_i)$ to denote the imaginary part. Let

$$(3.17) \quad g(X_1, X_2, \dots, X_l) = \sum_{\alpha \in G} X_\alpha - \Re(\Theta_i) \sim 0$$

and

$$(3.18) \quad h(Y_1, Y_2, \dots, Y_l) = \sum_{\alpha \in G} Y_\alpha - \Im(\Theta_i) \sim 0.$$

We define

$$\begin{aligned} L(X_1, X_2, \dots, X_l, Y_1, Y_2, \dots, Y_l, s, t) &= f(X_1, X_2, \dots, X_l, Y_1, Y_2, \dots, Y_l) \\ &\quad + sg(X_1, X_2, \dots, X_l) + th(Y_1, Y_2, \dots, Y_l) \\ &= \sum_{\alpha \in G} \frac{\lambda_\alpha^2 - 1}{\lambda_\alpha} (X_\alpha^2 + Y_\alpha^2) \\ &\quad + s \sum_{\alpha \in G} X_\alpha - s\Re(\Theta_i) + t \sum_{\alpha \in G} Y_\alpha - t\Im(\Theta_i). \end{aligned}$$

Taking partial derivatives for L , we have, for $\alpha = 1, 2, \dots, l$,

$$\begin{aligned} \frac{\partial L}{\partial X_\alpha} &= 2 \frac{\lambda_\alpha^2 - 1}{\lambda_\alpha} X_\alpha + s, \\ \frac{\partial L}{\partial Y_\alpha} &= 2 \frac{\lambda_\alpha^2 - 1}{\lambda_\alpha} Y_\alpha + t, \\ \frac{\partial L}{\partial s} &= \sum_{\alpha \in G} X_\alpha - \Re(\Theta_i), \end{aligned}$$

and

$$\frac{\partial L}{\partial t} = \sum_{\alpha \in G} Y_\alpha - \Im(\Theta_i).$$

We denote $\theta_\alpha = \arctan \lambda_\alpha$ for $\alpha = 1, 2, \dots, l$, $\lambda_\alpha = \tan \theta_\alpha$ and $\Theta \sim \sum_{\alpha=1}^l \theta_\alpha$. Since $\Theta \in (0, \frac{\pi}{2})$, it is clear that only one θ_1 can belong to $[\frac{\pi}{4}, \frac{\pi}{2})$. To achieve our goal, we need to analyze three cases: $\theta_1 \in (0, \frac{\pi}{4})$, $\theta_1 = \frac{\pi}{4}$ and $\theta_1 \in (\frac{\pi}{4}, \frac{\pi}{2})$.

Case 1. $\theta_1 \in (0, \frac{\pi}{4})$. This implies that for any $\alpha = 1, 2, \dots, l$, we have $\lambda_\alpha \in (0, 1)$.

Let

$$\frac{\partial L}{\partial X_\alpha} = 0, \quad \frac{\partial L}{\partial Y_\alpha} = 0, \quad \alpha = 1, 2, \dots, l, \quad \frac{\partial L}{\partial s} \sim 0, \quad \text{and} \quad \frac{\partial L}{\partial t} \sim 0,$$

we get

$$X_\alpha \sim \left(\sum_{\beta \in G} \frac{\lambda_\beta}{\lambda_\beta^2 - 1} \right)^{-1} \frac{\lambda_\alpha}{\lambda_\alpha^2 - 1} \Re(\Theta_i),$$

and

$$Y_\alpha \sim \left(\sum_{\beta \in G} \frac{\lambda_\beta}{\lambda_\beta^2 - 1} \right)^{-1} \frac{\lambda_\alpha}{\lambda_\alpha^2 - 1} \Im(\Theta_i).$$

Thus, the point

$$(3.19) \quad \left(\sum_{\beta \in G} \frac{\lambda_\beta}{\lambda_\beta^2 - 1} \right)^{-1} \left(\frac{\lambda_\alpha}{\lambda_\alpha^2 - 1} \Re(\Theta_i), \frac{\lambda_\alpha}{\lambda_\alpha^2 - 1} \Im(\Theta_i) \right)_{\alpha=1,2,\dots,l}$$

is a stationary point of the quadratic polynomial f (see (3.16)) subject to the constraints (3.17) and (3.18).

We now show that this point (3.19) is in fact the maximizer of f under these constraints. To demonstrate this, we reformulate the problem as an unconstrained optimization problem. By (3.17) and (3.18), we get

$$(3.20) \quad \begin{aligned} X_1^2 + Y_1^2 &\sim \left(\Re(\Theta_i) - \sum_{\alpha=2}^l X_\alpha \right)^2 + \left(\Im(\Theta_i) - \sum_{\alpha=2}^l Y_\alpha \right)^2 \\ &= |\Theta_i|^2 - 2\Re(\Theta_i) \sum_{\alpha=2}^l X_\alpha - 2\Im(\Theta_i) \sum_{\alpha=2}^l Y_\alpha + \left(\sum_{\alpha=2}^l X_\alpha \right)^2 + \left(\sum_{\alpha=2}^l Y_\alpha \right)^2 \end{aligned}$$

Substituting (3.20) into (3.16), we obtain

$$\begin{aligned} &f(X_2, X_3, \dots, X_l, Y_2, Y_3, \dots, Y_l) \\ &\sim \sum_{\alpha=2}^l \left(\frac{\lambda_\alpha^2 - 1}{\lambda_\alpha} + \frac{\lambda_1^2 - 1}{\lambda_1} \right) (X_\alpha^2 + Y_\alpha^2) + \frac{\lambda_1^2 - 1}{\lambda_1} \sum_{\substack{\alpha \neq \beta \\ \alpha, \beta=2,3,\dots,l}} X_\alpha X_\beta + \frac{\lambda_1^2 - 1}{\lambda_1} \sum_{\substack{\alpha \neq \beta \\ \alpha, \beta=2,3,\dots,l}} Y_\alpha Y_\beta \\ &\quad + \frac{\lambda_1^2 - 1}{\lambda_1} |\Theta_i|^2 - 2\Re(\Theta_i) \frac{\lambda_1^2 - 1}{\lambda_1} \sum_{\alpha=2}^l X_\alpha - 2\Im(\Theta_i) \frac{\lambda_1^2 - 1}{\lambda_1} \sum_{\alpha=2}^l Y_\alpha. \end{aligned}$$

So, the Hessian matrix of $f(X_2, X_3, \dots, X_l, Y_2, Y_3, \dots, Y_l)$ is

$$(3.21) \quad \begin{pmatrix} \frac{\partial^2 f}{\partial X_\alpha \partial X_\beta} & O \\ O & \frac{\partial^2 f}{\partial Y_\alpha \partial Y_\beta} \end{pmatrix}_{\alpha, \beta=2,3,\dots,l},$$

where

$$\left(\frac{\partial^2 f}{\partial X_\alpha \partial X_\beta} \right)_{\alpha, \beta=2,3,\dots,l} = \left(\frac{\partial^2 f}{\partial Y_\alpha \partial Y_\beta} \right)_{\alpha, \beta=2,3,\dots,l}$$

$$= \begin{pmatrix} 2\left(\frac{\lambda_2^2-1}{\lambda_2} + \frac{\lambda_1^2-1}{\lambda_1}\right) & \frac{\lambda_1^2-1}{\lambda_1} & \cdots & \frac{\lambda_1^2-1}{\lambda_1} & \frac{\lambda_1^2-1}{\lambda_1} \\ \frac{\lambda_1^2-1}{\lambda_1} & 2\left(\frac{\lambda_3^2-1}{\lambda_3} + \frac{\lambda_1^2-1}{\lambda_1}\right) & \cdots & \frac{\lambda_1^2-1}{\lambda_1} & \frac{\lambda_1^2-1}{\lambda_1} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ \frac{\lambda_1^2-1}{\lambda_1} & \frac{\lambda_1^2-1}{\lambda_1} & \cdots & 2\left(\frac{\lambda_{l-1}^2-1}{\lambda_{l-1}} + \frac{\lambda_1^2-1}{\lambda_1}\right) & \frac{\lambda_1^2-1}{\lambda_1} \\ \frac{\lambda_1^2-1}{\lambda_1} & \frac{\lambda_1^2-1}{\lambda_1} & \cdots & \frac{\lambda_1^2-1}{\lambda_1} & 2\left(\frac{\lambda_l^2-1}{\lambda_l} + \frac{\lambda_1^2-1}{\lambda_1}\right) \end{pmatrix}$$

and O is zero matrix. To prove that (3.16) reaches its maximum at the point (3.19) under the conditions (3.17) and (3.18), we only need to show that the matrix (3.21) is negative definite. Equivalently, it suffices to verify that the matrix

$$-\left(\frac{\partial^2 f}{\partial X_\alpha \partial X_\beta}\right)_{\alpha, \beta=2,3,\dots,l} = \frac{\lambda_1^2-1}{\lambda_1} \begin{pmatrix} 2\left(\frac{\lambda_1}{\lambda_1^2-1} \cdot \frac{1-\lambda_2^2}{\lambda_2} - 1\right) & -1 & \cdots & -1 \\ -1 & 2\left(\frac{\lambda_1}{\lambda_1^2-1} \cdot \frac{1-\lambda_3^2}{\lambda_3} - 1\right) & \cdots & -1 \\ \vdots & \vdots & \ddots & \vdots \\ -1 & -1 & \cdots & 2\left(\frac{\lambda_1}{\lambda_1^2-1} \cdot \frac{1-\lambda_l^2}{\lambda_l} - 1\right) \end{pmatrix}$$

is positive definite. We denote the matrix by

$$(3.22) \quad M = \begin{pmatrix} 2\left(\frac{\lambda_1}{\lambda_1^2-1} \cdot \frac{1-\lambda_2^2}{\lambda_2} - 1\right) & -1 & \cdots & -1 \\ -1 & 2\left(\frac{\lambda_1}{\lambda_1^2-1} \cdot \frac{1-\lambda_3^2}{\lambda_3} - 1\right) & \cdots & -1 \\ \vdots & \vdots & \ddots & \vdots \\ -1 & -1 & \cdots & 2\left(\frac{\lambda_1}{\lambda_1^2-1} \cdot \frac{1-\lambda_l^2}{\lambda_l} - 1\right) \end{pmatrix}.$$

Since $\theta_1 \in (0, \frac{\pi}{4})$, we have $\frac{\lambda_1^2-1}{\lambda_1} < 0$. Thus, we need to show that the matrix

$$M' = -M = \begin{pmatrix} 2\left(\frac{\lambda_1}{\lambda_1^2-1} \cdot \frac{\lambda_2^2-1}{\lambda_2} + 1\right) & 1 & \cdots & 1 \\ 1 & 2\left(\frac{\lambda_1}{\lambda_1^2-1} \cdot \frac{\lambda_3^2-1}{\lambda_3} + 1\right) & \cdots & 1 \\ \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & \cdots & 2\left(\frac{\lambda_1}{\lambda_1^2-1} \cdot \frac{\lambda_l^2-1}{\lambda_l} + 1\right) \end{pmatrix}_{l-1 \times l-1}$$

is positive definite, which means checking that all $k-1$ -th order leading principal minors $\det(M'_{k-1})$ of the matrix M' are positive, where $k = 2, 3, \dots, l$. It follows immediately

from Lemma 2.5 that

$$\begin{aligned} \det(M'_{k-1}) &= \prod_{i=2}^k \left(2 \left(\frac{\lambda_1}{\lambda_1^2 - 1} \cdot \frac{\lambda_i^2 - 1}{\lambda_i} + 1 \right) - 1 \right) + \sum_{i=2}^k \prod_{\substack{j \neq i \\ j=2,3,\dots,k}} \left(2 \left(\frac{\lambda_1}{\lambda_1^2 - 1} \cdot \frac{\lambda_j^2 - 1}{\lambda_j} + 1 \right) - 1 \right) \\ &= \prod_{i=2}^k \left(2 \frac{\lambda_1}{\lambda_1^2 - 1} \cdot \frac{\lambda_i^2 - 1}{\lambda_i} + 1 \right) + \sum_{i=2}^k \prod_{\substack{j \neq i \\ j=2,3,\dots,k}} \left(2 \frac{\lambda_1}{\lambda_1^2 - 1} \cdot \frac{\lambda_j^2 - 1}{\lambda_j} + 1 \right). \end{aligned}$$

In this case, since $\lambda_\alpha \in (0, 1)$ for $\alpha = 1, 2, \dots, l$, we clearly have $\det(M'_{k-1}) > 0$ for all $k = 2, 3, \dots, l$.

With this, we have shown that

$$(3.23) \quad f(X_1, X_2, \dots, X_l, Y_1, Y_2, \dots, Y_l) \lesssim \left(\sum_{\alpha \in G} \frac{\lambda_\alpha}{\lambda_\alpha^2 - 1} \right)^{-1} |\Theta_i|^2$$

under the conditions (3.17) and (3.18). Thus, putting (3.23) into (3.15), we get

$$(3.24) \quad \frac{1}{\sigma_l(G)} \sum_{\alpha} F^{\alpha\bar{\alpha}} \phi_{\alpha\bar{\alpha}} \lesssim \sum_{i \in B} \left(\left(\sum_{\alpha \in G} \frac{\lambda_\alpha}{\lambda_\alpha^2 - 1} \right)^{-1} |\Theta_i|^2 + \Theta_{i\bar{i}} \right).$$

To obtain (3.1), we state the following claim.

Claim 1: If $\Theta \in (0, \frac{\pi}{2})$ and $\theta_1 \in (0, \frac{\pi}{4})$, then

$$(3.25) \quad \left(\sum_{\alpha \in G} \frac{\lambda_\alpha}{\lambda_\alpha^2 - 1} \right)^{-1} \leq \tan \Theta - \frac{1}{\tan \Theta},$$

where $G = \{1, 2, \dots, l\}$.

Proof of Claim 1. To prove this claim, we further divide the proof into two subcases.

- $\Theta \in (0, \frac{\pi}{4})$. Noting that $\frac{\lambda_\alpha}{\lambda_\alpha^2 - 1} = -\frac{1}{2} \tan 2\theta_\alpha$, verifying (3.25) equivalent to verifying

$$\sum_{\alpha \in G} \tan 2\theta_\alpha \leq \tan 2\Theta.$$

Since $\Theta \in (0, \frac{\pi}{4})$, for any choice of θ_β and θ_γ , where $\beta, \gamma = 1, 2, \dots, l$, we have $\tan(2\theta_\beta + 2\theta_\gamma) > 0$, $\tan 2\theta_\beta > 0$ and $\tan 2\theta_\gamma > 0$. Therefore, $\tan(2\theta_\beta + 2\theta_\gamma) = \frac{\tan 2\theta_\beta + \tan 2\theta_\gamma}{1 - \tan 2\theta_\beta \tan 2\theta_\gamma}$ implies that

$$\tan 2\theta_\beta + \tan 2\theta_\gamma \leq \tan(2\theta_\beta + 2\theta_\gamma).$$

Following this, we can obtain

$$(3.26) \quad \sum_{\alpha \in G} \tan 2\theta_\alpha \leq \tan 2 \left(\sum_{\alpha \in G} \theta_\alpha \right) \leq \tan 2\Theta.$$

• $\Theta \in [\frac{\pi}{4}, \frac{\pi}{2})$. Observing that $\lambda_\alpha \in (0, 1)$ for any $\alpha = 1, 2, \dots, l$. So, the left-hand side of (3.25) is negative, while the right-hand side is non-negative since $\Theta \in [\frac{\pi}{4}, \frac{\pi}{2})$. Clearly, (3.25) holds.

We have completed the proof of claim. #

Substituting (3.25) into (3.24), we get

$$(3.27) \quad \frac{1}{\sigma_l(G)} \sum_{\alpha} F^{\alpha\bar{\alpha}} \phi_{\alpha\bar{\alpha}} \lesssim \sum_{i \in B} \left(\left(\tan \Theta - \frac{1}{\tan \Theta} \right) |\Theta_i|^2 + \Theta_{i\bar{i}} \right).$$

Since $-\log \tan \Theta$ is plurisubharmonic, the right-hand side of (3.27) is non-positive, and then

$$\sum_{\alpha} F^{\alpha\bar{\alpha}} \phi_{\alpha\bar{\alpha}} \lesssim 0.$$

In fact, $-\log \tan \Theta$ is plurisubharmonic means that

$$(3.28) \quad \sum_{i=1}^n (\log \tan \Theta)_{i\bar{i}} = \frac{\sec^2 \Theta}{\tan \Theta} \sum_{i=1}^n \left(\left(\tan \Theta - \frac{1}{\tan \Theta} \right) |\Theta_i|^2 + \Theta_{i\bar{i}} \right) \leq 0.$$

Hence, we finish the proof of Lemma 3.1 in the case that $\theta_1 \in (0, \frac{\pi}{4})$.

Case 2. $\theta_1 = \frac{\pi}{4}$. This implies that $\Theta \in [\frac{\pi}{4}, \frac{\pi}{2})$, and thus $\lambda_1 = 1$, $\lambda_\beta \in [0, 1)$ for $\beta = 2, 3, \dots, l$. Therefore, from (3.15), we conclude with

$$\frac{1}{\sigma_l(G)} \sum_{\alpha} F^{\alpha\bar{\alpha}} \phi_{\alpha\bar{\alpha}} \lesssim \sum_{i \in B} \Theta_{i\bar{i}}.$$

According to $\Theta \in [\frac{\pi}{4}, \frac{\pi}{2})$, we know that $\tan \Theta - \frac{1}{\tan \Theta} \geq 0$. Thus, as $-\log \tan \Theta$ is plurisubharmonic, and from (3.28), for any $i \in B = \{l+1, l+2, \dots, n\}$, we have $\Theta_{i\bar{i}} \leq 0$. Hence,

$$\sum_{\alpha} F^{\alpha\bar{\alpha}} \phi_{\alpha\bar{\alpha}} \lesssim 0.$$

Case 3. $\theta_1 \in (\frac{\pi}{4}, \frac{\pi}{2})$. Similar to **Case 1**, we still need to verify that the point (3.19) indeed yields the maximum of f (see (3.16)) under the constraints (3.17) and (3.18). Since $\lambda_1 = \tan \theta_1 > 1$, it suffices to prove that the matrix M (see (3.22)) is positive definite. This verification requires considerable technical care, and due to its complexity, the detailed proof is provided in **Appendix A**. Once this has been established, we obtain

$$\frac{1}{\sigma_l(G)} \sum_{\alpha} F^{\alpha\bar{\alpha}} \phi_{\alpha\bar{\alpha}} \lesssim \sum_{i \in B} \left(\left(\sum_{\alpha \in G} \frac{\lambda_{\alpha}}{\lambda_{\alpha}^2 - 1} \right)^{-1} |\Theta_i|^2 + \Theta_{i\bar{i}} \right).$$

Next, we need to derive (3.25) under the condition $\theta_1 \in (\frac{\pi}{4}, \frac{\pi}{2})$. In fact, we clearly have $\tan 2\theta_1 < 0$ and $\tan 2\Theta < 0$, and thus we get

$$\tan 2\Theta = \tan \left(2\theta_1 + 2 \sum_{\beta=2}^n \theta_{\beta} \right)$$

$$\begin{aligned}
&= \frac{\tan 2\theta_1 + \tan(2 \sum_{\beta=2}^n \theta_\beta)}{1 - \tan 2\theta_1 \tan(2 \sum_{\beta=2}^n \theta_\beta)} \\
&\geq \tan 2\theta_1 + \tan \left(2 \sum_{\beta=2}^n \theta_\beta \right).
\end{aligned}$$

Since $\sum_{\beta=2}^n \theta_\beta \in (0, \frac{\pi}{4})$, we can again apply **Claim 1** for the case $\Theta \in (0, \frac{\pi}{4})$, which gives (3.26), namely

$$\sum_{\alpha \in G} \tan 2\theta_\alpha \leq \tan 2\Theta.$$

Consequently, (3.25) holds, and this further yields

$$\frac{1}{\sigma_l(G)} \sum_{\alpha} F^{\alpha\bar{\alpha}} \phi_{\alpha\bar{\alpha}} \lesssim \sum_{i \in B} \left(\left(\tan \Theta - \frac{1}{\tan \Theta} \right) |\Theta_i|^2 + \Theta_{i\bar{i}} \right).$$

Again, based on $-\log \tan \Theta$ is plurisubharmonic, we arrive at

$$\sum_{\alpha} F^{\alpha\bar{\alpha}} \phi_{\alpha\bar{\alpha}} \lesssim 0.$$

Hence, we have completed the proof of Lemma 3.1. $\square$

At this point, we are ready to combine Lemma 3.1 with the strong maximum principle to establish the proof of Theorem 1.1.

Proof of Theorem 1.1. Denote

$$l = \min_{z \in \Omega} \text{rank}(\partial\bar{\partial}u(z)).$$

Assume that the complex Hessian $\partial\bar{\partial}u$ attains its minimal rank l at $z_0 \in \Omega$. We will prove that $\text{rank}(\partial\bar{\partial}u(z)) = l$ for all $z \in \Omega$. Define the set

$$\Omega^* = \{z \in \Omega : \text{rank}(\partial\bar{\partial}u(z)) = l\}.$$

It is clear that Ω^* is relatively closed in Ω . Combining the strong maximum principle and Lemma 3.1, we know that Ω^* is also open. Thus, $\Omega = \Omega^*$, i.e., the rank of complex Hessian $\partial\bar{\partial}u$ always l in Ω . $\square$

4. THE CASE OF KÄHLER MANIFOLDS

In this section, we investigate an analogous constant rank result on Kähler manifolds, namely Theorem 1.3. The proof closely follows the strategy used for Theorem 1.1, a notable distinction is the appearance of curvature terms when commuting covariant derivatives.

Proof of Theorem 1.3. We work on a small neighborhood $\mathcal{O}$ of $z_0 \in M$, which is a point where $W(z_0)$ is of minimum rank l . Set

$$\phi(z) = \sigma_{l+1}(w_{i\bar{j}}(z)).$$

For any point $z \in \mathcal{O}$, we choose a local orthonormal frame at z so that $W(z)$ is diagonal and $w_{i\bar{i}} = \lambda_i$ for any $i = 1, 2, \dots, n$. As in the proof of Lemma 3.1, we may divide eigenvalues of W at z into G and B , the “good” and “bad” sets of indices, respectively, and $|G| = l$ and $|B| = n - l$, where $|\cdot|$ stands for the number of elements in a set. The following calculations are still carried out at the fixed point z . Next, we use the notations introduced in the proof of Lemma 3.1. Our goal is to prove

$$(4.1) \quad \sum_{\alpha} F^{\alpha\bar{\alpha}} \phi_{\alpha\bar{\alpha}} \lesssim 0.$$

Once (4.1) is obtained, we can combine it with the strong maximum principle to finish the proof of Theorem 1.3.

Through similar calculations, we can obtain a formula similar to (3.7) derived in the proof of Lemma 3.1,

$$\begin{aligned} \sum_{\alpha} F^{\alpha\bar{\alpha}} \phi_{\alpha\bar{\alpha}} &\sim \sigma_l(G) \sum_{\alpha} \sum_{i \in B} F^{\alpha\bar{\alpha}} w_{i\bar{i}\alpha\bar{\alpha}} - \sigma_{l-1}(G) \sum_{\alpha} \sum_{i, j \in B} F^{\alpha\bar{\alpha}} |w_{i\bar{j}\alpha}|^2 \\ &\quad - \sum_{\alpha} \sum_{\substack{i \in G \\ j \in B}} \sigma_{l-1}(G|i) F^{\alpha\bar{\alpha}} (|w_{i\bar{j}\alpha}|^2 + |w_{j\bar{i}\alpha}|^2). \end{aligned}$$

Using the commutation formulas for covariant derivatives on Kähler manifolds, we have

$$\begin{aligned} w_{i\bar{i}\alpha\bar{\alpha}} &= w_{\alpha\bar{i}i\bar{\alpha}} \\ &= w_{\alpha\bar{i}\bar{\alpha}i} + \sum_m w_{m\bar{i}} R_{\alpha\bar{m}i\bar{\alpha}} - \sum_m w_{\alpha\bar{m}} R_{m\bar{i}i\bar{\alpha}} \\ &= w_{\alpha\bar{\alpha}i\bar{i}} + w_{i\bar{i}} R_{i\bar{i}\alpha\bar{\alpha}} - w_{\alpha\bar{\alpha}} R_{\alpha\bar{\alpha}i\bar{i}}. \end{aligned}$$

It follows that

$$\begin{aligned} \sum_{\alpha} F^{\alpha\bar{\alpha}} \phi_{\alpha\bar{\alpha}} &\sim \sigma_l(G) \sum_{\alpha} \sum_{i \in B} F^{\alpha\bar{\alpha}} w_{\alpha\bar{\alpha}i\bar{i}} - \sigma_{l-1}(G) \sum_{\alpha} \sum_{i, j \in B} F^{\alpha\bar{\alpha}} |w_{i\bar{j}\alpha}|^2 \\ (4.2) \quad &\quad - \sum_{\alpha} \sum_{\substack{i \in G \\ j \in B}} \sigma_{l-1}(G|i) F^{\alpha\bar{\alpha}} (|w_{i\bar{j}\alpha}|^2 + |w_{j\bar{i}\alpha}|^2) \\ &\quad - \sigma_l(G) \sum_{\alpha \in G} \sum_{i \in B} F^{\alpha\bar{\alpha}} w_{\alpha\bar{\alpha}} R_{\alpha\bar{\alpha}i\bar{i}}. \end{aligned}$$

We denote $\theta_{\alpha} = \arctan \lambda_{\alpha}$ for $\alpha = 1, 2, \dots, l$, $\lambda_{\alpha} = \tan \theta_{\alpha}$ and $\Theta \sim \sum_{\alpha=1}^l \theta_{\alpha}$. When $\theta_1 \in (0, \frac{\pi}{4}) \cup (\frac{\pi}{4}, \frac{\pi}{2})$, by following the exact steps that led to (3.27) in Lemma 3.1, we can derive from (4.2) that

$$(4.3) \quad \frac{1}{\sigma_l(G)} \sum_{\alpha} F^{\alpha\bar{\alpha}} \phi_{\alpha\bar{\alpha}} \lesssim \sum_{i \in B} \left(\left(\tan \Theta - \frac{1}{\tan \Theta} \right) |\Theta_i|^2 + \Theta_{i\bar{i}} \right) - r(n-l) \sum_{\alpha \in G} F^{\alpha\bar{\alpha}} w_{\alpha\bar{\alpha}}.$$

Similarly to **Case 2** in Lemma 3.1, when $\theta_1 = \frac{\pi}{4}$, we can obtain

$$(4.4) \quad \begin{aligned} \frac{1}{\sigma_l(G)} \sum_{\alpha} F^{\alpha\bar{\alpha}} \phi_{\alpha\bar{\alpha}} &\lesssim \sum_{i \in B} \Theta_{i\bar{i}} - r \sum_{\alpha \in G} F^{\alpha\bar{\alpha}} w_{\alpha\bar{\alpha}} \\ &\leq \sum_{i \in B} \left(\left(\tan \Theta - \frac{1}{\tan \Theta} \right) |\Theta_i|^2 + \Theta_{i\bar{i}} \right) - r(n-l) \sum_{\alpha \in G} F^{\alpha\bar{\alpha}} w_{\alpha\bar{\alpha}}, \end{aligned}$$

since $\Theta \in [\frac{\pi}{4}, \frac{\pi}{2})$. For the last terms in (4.3) or (4.4), we have

$$\sum_{\alpha \in G} F^{\alpha\bar{\alpha}} w_{\alpha\bar{\alpha}} = \sum_{\alpha=1}^l \frac{\tan \theta_{\alpha}}{1 + \tan^2 \theta_{\alpha}} = \frac{1}{2} \sum_{\alpha=1}^l \sin 2\theta_{\alpha}.$$

Noting that $\Theta \in (0, \frac{\pi}{2})$ and $\theta_{\alpha} > 0$ for any $\alpha = 1, 2, \dots, l$, by using induction, we arrive at

$$\begin{aligned} \frac{1}{2} \sin \left(2 \sum_{\alpha=1}^l \theta_{\alpha} \right) &= \frac{1}{2} \sin \left(2\theta_1 + 2 \sum_{\alpha=2}^l \theta_{\alpha} \right) \\ &= \frac{1}{2} \sin 2\theta_1 \cos \left(2 \sum_{\alpha=2}^l \theta_{\alpha} \right) + \frac{1}{2} \cos 2\theta_1 \sin \left(2 \sum_{\alpha=2}^l \theta_{\alpha} \right) \\ &\leq \frac{1}{2} \sin 2\theta_1 + \frac{1}{2} \sin \left(2\theta_2 + 2 \sum_{\alpha=3}^l \theta_{\alpha} \right) \\ &\leq \dots \\ &\leq \frac{1}{2} \sum_{\alpha=1}^l \sin 2\theta_{\alpha}. \end{aligned}$$

Since $\frac{1}{2} \sin(2 \sum_{\alpha=1}^l \theta_{\alpha}) \sim \frac{\tan \Theta}{\sec^2 \Theta}$, it follows that

$$(4.5) \quad \frac{\tan \Theta}{\sec^2 \Theta} \lesssim \sum_{\alpha \in G} F^{\alpha\bar{\alpha}} w_{\alpha\bar{\alpha}}.$$

Observing that

$$(\log \tan \Theta)_{i\bar{i}} = \frac{\sec^2 \Theta}{\tan \Theta} \left(\left(\tan \Theta - \frac{1}{\tan \Theta} \right) |\Theta_i|^2 + \Theta_{i\bar{i}} \right),$$

and combining (4.3) or (4.4), and (4.5), we conclude with

$$\frac{1}{\sigma_l(G)} \sum_{\alpha} F^{\alpha\bar{\alpha}} \phi_{\alpha\bar{\alpha}} \lesssim \frac{\tan \Theta}{\sec^2 \Theta} \left(\sum_{i \in B} (\log \tan \Theta)_{i\bar{i}} - r(n-l) \right).$$

According to our assumptions, $(\log \tan \Theta)_{i\bar{j}} - r g_{i\bar{j}}$ is non-positive definite, so

$$\sum_{\alpha} F^{\alpha\bar{\alpha}} \phi_{\alpha\bar{\alpha}} \lesssim 0.$$

Hence, we finish the proof of Theorem 1.3. □

APPENDIX A. PROOF OF CASE 3 IN LEMMA 3.1

In the appendix, we complete the proof of **Case 3** in Lemma 3.1 by verifying that the matrix

$$M = \begin{pmatrix} 2\left(\frac{\lambda_1}{\lambda_1^2-1} \cdot \frac{1-\lambda_2^2}{\lambda_2} - 1\right) & -1 & \cdots & -1 \\ -1 & 2\left(\frac{\lambda_1}{\lambda_1^2-1} \cdot \frac{1-\lambda_3^2}{\lambda_3} - 1\right) & \cdots & -1 \\ \vdots & \vdots & \ddots & \vdots \\ -1 & -1 & \cdots & 2\left(\frac{\lambda_1}{\lambda_1^2-1} \cdot \frac{1-\lambda_l^2}{\lambda_l} - 1\right) \end{pmatrix}_{l-1 \times l-1}$$

is positive definite, given that $\lambda_1 > 1$ and $\lambda_2, \lambda_3, \dots, \lambda_l \in (0, 1)$.

Proof of Case 3 in Lemma 3.1. It only requires checking that all $k-1$ -th order leading principal minors of M are positive for $k = 2, 3, \dots, l$.

According to Lemma 2.5, we get that the $k-1$ -th order leading principal minors $\det(M_{k-1})$ of the matrix M is

$$\begin{aligned} \det(M_{k-1}) &= \prod_{i=2}^k \left(2\left(\frac{\lambda_1}{\lambda_1^2-1} \cdot \frac{1-\lambda_i^2}{\lambda_i} - 1\right) + 1 \right) - \sum_{i=2}^k \prod_{\substack{j \neq i \\ j=2,3,\dots,k}} \left(2\left(\frac{\lambda_1}{\lambda_1^2-1} \cdot \frac{1-\lambda_j^2}{\lambda_j} - 1\right) + 1 \right) \\ &= \prod_{i=2}^k \left(2\frac{\lambda_1}{\lambda_1^2-1} \cdot \frac{1-\lambda_i^2}{\lambda_i} - 1 \right) - \sum_{i=2}^k \prod_{\substack{j \neq i \\ j=2,3,\dots,k}} \left(2\frac{\lambda_1}{\lambda_1^2-1} \cdot \frac{1-\lambda_j^2}{\lambda_j} - 1 \right). \end{aligned}$$

Noting that $\frac{\lambda_i}{\lambda_i^2-1} = -\frac{1}{2} \tan 2\theta_i$, $\det(M_{k-1})$ can be written as

$$\det(M_{k-1}) = (-1)^{k-1} \prod_{i=2}^k \left(2 \frac{\tan 2\theta_1}{\tan 2\theta_i} + 1 \right) + (-1)^{k-1} \sum_{i=2}^k \prod_{\substack{j \neq i \\ j=2,3,\dots,k}} \left(2 \frac{\tan 2\theta_1}{\tan 2\theta_j} + 1 \right),$$

and

(A.1)

$$\det(M_{k-1}) = \frac{(-1)^{k-1}}{\prod_{i=2}^k \tan 2\theta_i} \left(\prod_{i=2}^k (2 \tan 2\theta_1 + \tan 2\theta_i) + \sum_{i=2}^k \left(\prod_{j \neq i} (2 \tan 2\theta_1 + \tan 2\theta_j) \tan 2\theta_i \right) \right).$$

Let $\mu_i = \tan 2\theta_i$, and the vector $\mu = (\mu_2, \mu_3, \dots, \mu_k)$. Using the notation of elementary symmetric functions, we can write (A.1) as

(A.2)

$$\begin{aligned} \det(M_{k-1}) &= \frac{(-1)^{k-1}}{\sigma_{k-1}} \left(\sum_{p=0}^{k-1} (2 \tan 2\theta_1)^p \sigma_{k-1-p} + \sum_{p=0}^{k-2} (k-1-p)(2 \tan 2\theta_1)^p \sigma_{k-1-p} \right) \\ &= \frac{(-1)^{k-1}}{\sigma_{k-1}} \sum_{p=0}^{k-1} (k-p)(2 \tan 2\theta_1)^p \sigma_{k-1-p}. \end{aligned}$$

To show that $\det(M_{k-1}) > 0$, we will consider the case where $k-1$ is odd and even, expanding (A.2) and then regrouping.

- $k-1$ is odd.

$$\begin{aligned} \det(M_{k-1}) &= \frac{(-1)^{k-1}}{\sigma_{k-1}} \left((2 \tan 2\theta_1)^{k-1} + 2(2 \tan 2\theta_1)^{k-2} \sigma_1 + 3(2 \tan 2\theta_1)^{k-3} \sigma_2 \right. \\ &\quad \left. + 4(2 \tan 2\theta_1)^{k-4} \sigma_3 + \dots + (k-3)(2 \tan 2\theta_1)^3 \sigma_{k-4} \right. \\ &\quad \left. + (k-2)(2 \tan 2\theta_1)^2 \sigma_{k-3} + (k-1)(2 \tan 2\theta_1) \sigma_{k-2} + k \sigma_{k-1} \right) \\ &= \frac{(-1)^{k-1}}{\sigma_{k-1}} \left(2(2 \tan 2\theta_1)^{k-2} (\tan 2\theta_1 \sigma_0 + \sigma_1) \right. \\ &\quad \left. + 4(2 \tan 2\theta_1)^{k-4} (\tan 2\theta_1 \sigma_2 + \sigma_3) + 2 \tan 2\theta_1 (2 \tan 2\theta_1)^{k-4} \sigma_2 \right. \\ &\quad \left. + 6(2 \tan 2\theta_1)^{k-6} (\tan 2\theta_1 \sigma_4 + \sigma_5) + 4 \tan 2\theta_1 (2 \tan 2\theta_1)^{k-6} \sigma_4 \right. \\ &\quad \left. + \dots \right. \\ &\quad \left. + (k-2)(2 \tan 2\theta_1)^2 (\tan 2\theta_1 \sigma_{k-4} + \sigma_{k-3}) + (k-4) \tan 2\theta_1 (2 \tan 2\theta_1)^2 \sigma_{k-4} \right. \\ &\quad \left. + k(\tan 2\theta_1 \sigma_{k-2} + \sigma_{k-1}) + (k-2) \tan 2\theta_1 \sigma_{k-2} \right) \\ &= \frac{(-1)^{k-1}}{\sigma_{k-1}} \sum_{q=0}^{\frac{k-2}{2}} \left(2(q+1)(2 \tan 2\theta_1)^{k-2(q+1)} (\tan 2\theta_1 \sigma_{2q} + \sigma_{2q+1}) \right. \\ &\quad \left. + q(2 \tan 2\theta_1)^{k-2(q+1)+1} \sigma_{2q} \right). \end{aligned}$$

When $k-1$ is odd, $\tan 2\theta_1 < 0$ by $\theta_1 \in (\frac{\pi}{4}, \frac{\pi}{2})$, it implies that for any $q = 0, 1, \dots, \frac{k-2}{2}$, we have

$$\frac{(-1)^{k-1}}{\sigma_{k-1}} q(2 \tan 2\theta_1)^{k-2(q+1)+1} \sigma_{2q} \geq 0,$$

hence

$$(A.3) \quad \det(M_{k-1}) \geq \frac{(-1)^{k-1}}{\sigma_{k-1}} \sum_{q=0}^{\frac{k-2}{2}} 2(q+1)(2 \tan 2\theta_1)^{k-2(q+1)} (\tan 2\theta_1 \sigma_{2q} + \sigma_{2q+1}).$$

• $k-1$ is even.

$$\begin{aligned} \det(M_{k-1}) &= \frac{(-1)^{k-1}}{\sigma_{k-1}} \left(2(2 \tan 2\theta_1)^{k-2} (\tan 2\theta_1 \sigma_0 + \sigma_1) \right. \\ &\quad + 4(2 \tan 2\theta_1)^{k-4} (\tan 2\theta_1 \sigma_2 + \sigma_3) + 2 \tan 2\theta_1 (2 \tan 2\theta_1)^{k-4} \sigma_2 \\ &\quad + 6(2 \tan 2\theta_1)^{k-6} (\tan 2\theta_1 \sigma_4 + \sigma_5) + 4 \tan 2\theta_1 (2 \tan 2\theta_1)^{k-6} \sigma_4 \\ &\quad + \dots \\ &\quad + (k-1)(2 \tan 2\theta_1) (\tan 2\theta_1 \sigma_{k-3} + \sigma_{k-2}) + (k-3) \tan 2\theta_1 (2 \tan 2\theta_1) \sigma_{k-3} \\ &\quad \left. + k \sigma_{k-1} \right) \\ &= \frac{(-1)^{k-1}}{\sigma_{k-1}} \sum_{q=0}^{\frac{k-3}{2}} \left(2(q+1)(2 \tan 2\theta_1)^{k-2(q+1)} (\tan 2\theta_1 \sigma_{2q} + \sigma_{2q+1}) \right. \\ &\quad \left. + q(2 \tan 2\theta_1)^{k-2(q+1)+1} \sigma_{2q} \right) + (-1)^{k-1} k. \end{aligned}$$

Since $k-1$ is even, for any $q = 0, 1, \dots, \frac{k-3}{2}$, we have

$$\frac{(-1)^{k-1}}{\sigma_{k-1}} q(2 \tan 2\theta_1)^{k-2(q+1)+1} \sigma_{2q} \geq 0$$

and $(-1)^{k-1} k > 0$, hence

$$(A.4) \quad \det(M_{k-1}) \geq \frac{(-1)^{k-1}}{\sigma_{k-1}} \sum_{q=0}^{\frac{k-3}{2}} 2(q+1)(2 \tan 2\theta_1)^{k-2(q+1)} (\tan 2\theta_1 \sigma_{2q} + \sigma_{2q+1}).$$

For (A.3) and (A.4), by using the floor function $\lfloor \cdot \rfloor$, we can unify them as

$$(A.5) \quad \det(M_{k-1}) \geq \frac{(-1)^{k-1}}{\sigma_{k-1}} \sum_{q=0}^{\lfloor \frac{k-2}{2} \rfloor} 2(q+1)(2 \tan 2\theta_1)^{k-2(q+1)} (\tan 2\theta_1 \sigma_{2q} + \sigma_{2q+1}).$$

We note that in (A.5), the parity of $k-1$ and $k-2(q+1)$ are different, which implies that

$$\frac{(-1)^{k-1}}{\sigma_{k-1}} 2(q+1)(2 \tan 2\theta_1)^{k-2(q+1)} < 0,$$

for any $q = 0, 1, \dots, \lfloor \frac{k-2}{2} \rfloor$. Therefore, we only need to prove that

$$(A.6) \quad \tan 2\theta_1 \sigma_{2q} + \sigma_{2q+1} < 0$$

holds.

Next, we will verify (A.6). Since for any $i = 2, 3, \dots, k$, $\tan 2\theta_i \geq 0$, by the Newton's inequality, i.e., Lemma 2.2, we get

$$\frac{\sigma_{2q+1}}{\sigma_{2q}} \leq \frac{\sigma_{2q}}{\sigma_{2q-1}} \leq \dots \leq \frac{\sigma_2}{\sigma_1} \leq \frac{\sigma_1}{\sigma_0}.$$

It follows that

$$\tan 2\theta_1 + \frac{\sigma_{2q+1}}{\sigma_{2q}} \leq \tan 2\theta_1 + \sigma_1 = \sum_{i=1}^k \tan 2\theta_i.$$

In this case, where $\theta_1 \in (\frac{\pi}{4}, \frac{\pi}{2})$, it yields $\Theta \in (\frac{\pi}{4}, \frac{\pi}{2})$. Clearly, we can see that $\tan 2\theta_1 < 0$ and $\tan 2\Theta < 0$. Thus, we obtain

$$\begin{aligned} \tan 2\Theta &= \tan \left(2\theta_1 + 2 \sum_{i=2}^n \theta_i \right) \\ &= \frac{\tan 2\theta_1 + \tan(2 \sum_{i=2}^n \theta_i)}{1 - \tan 2\theta_1 \tan(2 \sum_{i=2}^n \theta_i)} \\ &\geq \tan 2\theta_1 + \tan \left(2 \sum_{i=2}^n \theta_i \right). \end{aligned}$$

Since $\sum_{i=2}^n \theta_i \in (0, \frac{\pi}{4})$, we apply (3.26) from **Case 1** in the proof of Lemma 3.1 and have

$$\sum_{i=1}^k \tan 2\theta_i \leq \tan(2\Theta).$$

According to $\tan 2\Theta < 0$, we then conclude that

$$\tan 2\theta_1 + \frac{\sigma_{2q+1}}{\sigma_{2q}} < 0,$$

thereby completing the proof of (A.6). $\square$

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